

Cryptocurrency Transaction Analysis

A Project Report Submitted in Partial Fulfillment for the certificate of
summer training of Bachelor of Technology in Computer Science and
Engineering



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PRAYAGRAJ,**

August, 2025

Certificate

This is to certify that the project titled “ **Cryptocurrency Transaction Analysis** ” submitted by **Aru Shubham Singh** (Reg. No. 2400100100114), **Ashish Giri** (Reg. No. 2400100100122), and **Avjeet Kumar** (Reg. No. 2400100100137) in partial fulfillment of the requirement for the certificate for Summer Training submitted at United College of Engineering & Research, Prayagraj is an authentic record of their own work carried out during a period from **July 2025 to August 2025** under the guidance of **Dr. Snehlata**, Assistant Professor Department of Computer Science and Engineering.

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Place: Prayagraj

Dated: 20th August 2025

Candidates Declaration

We hereby declare that our work submitted in the partial fulfilment of the requirements for the certificate of the Summer Training in the Department of Computer Science and Engineering of the United College of Research & Engineering, titled as “*Cryptocurrency Transaction Analysis*” is a record of our work carried out during the *Summer Vacation* from **July 2025 to August 2025** under the guidance of **Dr. Snehleta**, Assistant Professor, UCER.

The matter presented in this report has not been submitted by us for the award of any other degree of this or any other Institute/University.

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Acknowledgment

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Abstract

Cryptocurrency transaction analysis explores the methods and techniques to examine blockchain transaction data to understand patterns, detect anomalies, and ensure security. This field leverages advanced data analytics, machine learning, and graph theory to trace transaction flows, identify illicit activities such as money laundering, and assess market behavior. The decentralized and pseudonymous nature of cryptocurrencies presents unique challenges, requiring innovative analytical models to decode transaction linkages and user behaviors accurately. By analyzing transaction metadata, timestamps, and network structures, researchers can uncover hidden relationships, optimize transaction verification, and enhance regulatory compliance.

This abstract presents a comprehensive overview of state-of-the-art techniques applied to cryptocurrency transaction analysis, emphasizing the importance of transparency and trust in distributed ledger technologies. Further, it highlights future directions for enhancing scalability and privacy preservation while mitigating fraudulent manipulations within blockchain networks, thereby contributing to the sustainable growth and adoption of digital currencies.

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Chapter 1

Cryptocurrency

1.1 What is cryptocurrency?

A cryptocurrency is a digital or virtual currency protected by cryptography which makes counterfeiting or double-spending almost impossible. Most cryptocurrencies are decentralized, blockchain-based networks a public database operated by a dispersed computing network. One distinguishing characteristic of cryptocurrencies is that they are usually not distributed by any central agency, rendering them potentially resistant to intervention or abuse by the government.

The term "crypto-currency" derives from the encryption methods used to protect the network. Cryptocurrencies attract scrutiny for a variety of reasons including their use for illicit activity, exchange rate fluctuations, and network flows that underlie them. They were also praised for their portability, accountability and divisibility. Cryptocurrencies are almost always intended to be free of government influence and regulation, but this core feature of the technology has come under fire as they have become more common. The currencies modelled after bitcoin are called altcoins collectively and have often attempted to present themselves as modified or improved versions of bitcoin.

1.2 The top cryptocurrencies

Here are the top cryptocurrencies in India.

The cryptocurrency hype has traveled from the west and reached India, attracting investors



Figure 1.1: Cryptocurrencies

with its high value. Be it. seasoned investors or novice enthusiasts, everyone wants to partake in a conversation about cryptocurrencies and give opinions. While many NRIs have good knowledge about the growing cryptocurrency market. Indians need to update themselves with the market news.

After the recent market dip brought by China's crypto ban, the high crypto coin prices have caught the interest of Indian investors and skeptics. People, who were once supporting or investing in such a volatile market due to high prices, are now ready to make their first, albeit small cryptocurrency investment.

1. Bitcoin (BTC)

Price today: **INR 10,102,993.17**

For Indians, Bitcoin is synonymous with cryptocurrency. And rightfully so, because this was the first and is the highest valued crypto in the market right now. After reaching an all-time high of about **\$124000** at **14 august** this year, the price started plummeting recently. It is a matter of days of bitcoin. It increased by **100%** after 2024. Bitcoin's price increased significantly after 2023 due to a combination of factors, including regulatory developments, macroeconomic conditions, and renewed market interest.

2. Ethereum (ETH)

Price today: **INR 4,91,713.99**

Ether, introduced in 2015, is presently the second-biggest digital currency by market value behind bitcoin, but it lags by a substantial margin behind the dominant cryptocurrency. Ef-

fective August 14th, the market value of ether is around **1/26** the size of bitcoins. Ethereum is focused on realistic smart contracts for the digitalisation of transactions used by several companies. Ethereum is a decentralized computing framework that enables the construction and running of Smart Contracts and Decentralized Applications without any third party interruption, theft, power or intervention. On Ethereum, the programs ran on the platform-specific cryptographic token, ether.

3. Binance Coin (BNB)

Price today: **INR 66,218.06**

As per market capitalization, Binance Coin is the third-largest cryptocurrency, the first two being Bitcoin and Ethereum. In 2017, Binance Coin was launched by one of the world's largest cryptocurrency exchanger app **Binance**, as a utility token. Hence, the pricing of this crypto coin depends on its utility on the Binance platform. at **August 14, 2025, at \$868.41** is Its highest .

4. Ripple (XRP)

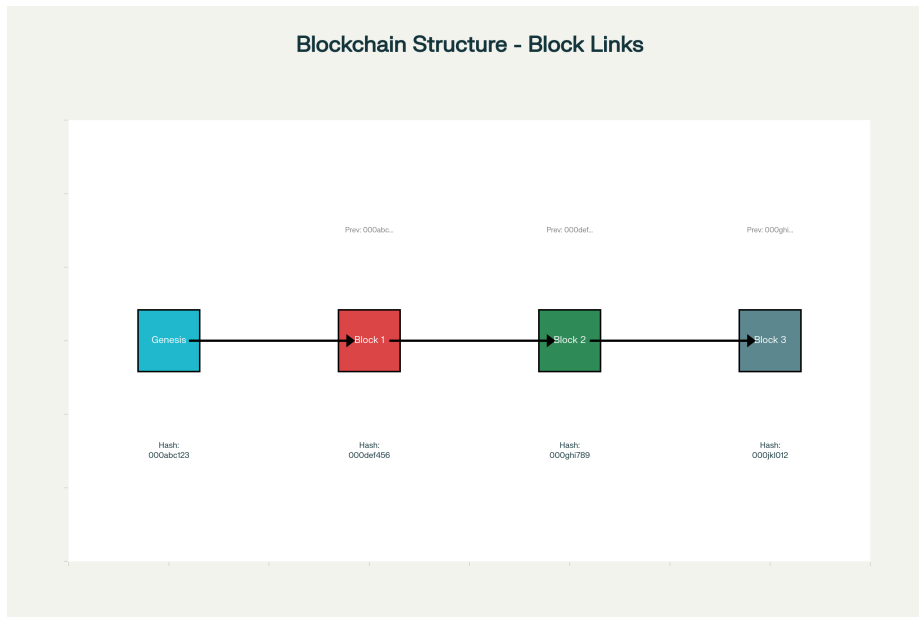
Price today : **INR 273.47**

Launched in 2012, Ripple helps banks to real-time settle cross-border trades for end-to-end transparency and lower costs. With its new business model, Ripple has seen success it remains one of the most appealing digital currencies. It is the coin where the transaction fees is very low. As my experience i always use this coin

There are a lot of coins launching.some coins like solana hamster,dog tom coin etc.

1.3 How does cryptocurrency work?

Cryptocurrencies run on a distributed public ledger called **blockchain technology**, a record of all transactions updated and held by currency holders.

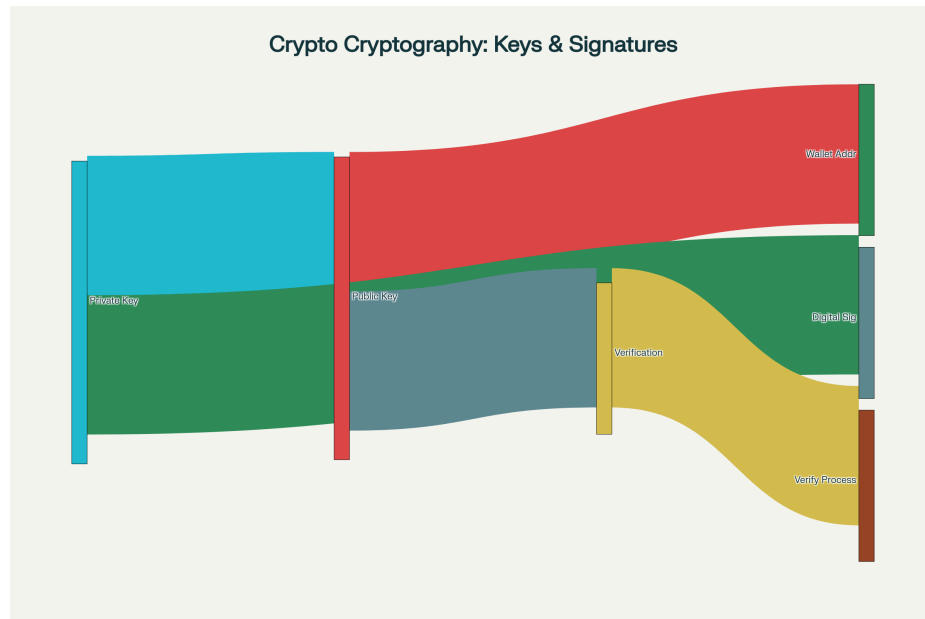


Units of cryptocurrency are created through a process called mining, which involves using computer power to solve complicated mathematical problems that generate coins. Users can also buy the currencies from brokers, then store and spend them using cryptographic wallets.

If you own cryptocurrency, you don't own anything tangible. What you own is a key that allows you to move a record or a unit of measure from one person to another without a trusted third party.

Although Bitcoin has been around since 2009, cryptocurrencies and applications of blockchain technology are still emerging financial terms, and more uses are expected in the future. Transactions including bonds, stocks, and other financial assets could eventually be traded using the technology

Here some graphs which shows the process.....



1.4 How To purchase a cryptocurrency

For purchasing any cryptocurrency follows these steps....

Step 1: Go to any verified cryptocurrency exchanger application like **Kucoin, Binance, CoinDCX etc..** . For my suggestion use Binance

Step 2: Create an account there by entering Email id and phone number . There is also need KYC.

Step 3: After Completing Kyc choose any currency from **Market Menu**

Before it add money by purchasing **USDT** by **P2P** method.

Step 4: After choosing coin, select **Buy** and select **Market price** (for instant purchase) and pay by USDT and done.

Step 5: Now you can track your coin from **Overview** menu.

1.5 TRANSACTION ANALYSIS

As our project, our main objective is to analysis the transaction of the cryptocurrency,

1.5.1 Transaction terms

Here are concise definitions of key cryptocurrency transaction terms:

1: Transaction (Tx): A transfer of cryptocurrency between addresses recorded on a blockchain.

2: Transaction Hash (TxID): A unique alphanumeric code identifying a specific transaction.

3: Sender Address: The wallet address that initiates the transfer of cryptocurrency.

4: Recipient Address: The wallet address receiving the cryptocurrency.

5: Input: The source of funds used in a transaction, referring to previous unspent outputs.

6: Output: The destination address(es) where the funds are sent in the transaction.

7: Transaction Fee: The amount paid to miners or validators to prioritize processing the transaction.

8: Block: A batch of transactions grouped and appended to the blockchain.

9: Confirmation: The process of validating and recording a transaction on the blockchain.

10: Unspent Transaction Output (UTXO): Outputs from previous transactions that are still available to be spent.

11: Double Spend: An attempt to spend the same cryptocurrency funds twice.

12: Signature: Cryptographic proof that senders ownership and authorize the transaction.

13: Timestamp: The recorded time when the transaction was created or confirmed.

14: Nonce: A number used to vary the transaction hash to find a valid proof-of-work.

15: Memo/Tag: Additional optional data in a transaction, often used for identification or notes.

1.5.2 How Transaction works

Here's how a cryptocurrency transaction works, step by step:

1. Initiation

The sender selects the amount of cryptocurrency to send and enters the recipient's wallet address in their wallet app.

The sender may also set the transaction fee: higher fees usually result in faster processing.

2. Signing

The sender's wallet cryptographically signs the transaction using their private key.

This signature proves ownership and authorization to send the coins.

3. Broadcasting

The signed transaction is broadcast to the blockchain network (nodes).

It becomes visible to all network participants.

4. Verification

Network nodes verify the transaction details:

Confirm the sender owns enough funds (unspent transaction outputs or account balance).

Validate the signature to ensure authenticity.

5. Transaction Pool

Validated transactions enter a pool (mempool), awaiting processing.

Miners (Proof-of-Work) or validators (Proof-of-Stake) select transactions from the pool.

6. Inclusion in a Block

The miner or validator adds the transaction to a new block, along with other pending transactions.

The block is then proposed to the network for consensus.

7. Confirmation and Finalization

Once the block is accepted and added to the blockchain, the transaction is considered confirmed.

Subsequent blocks layered on top provide additional confirmations, making the transaction increasingly secure and irreversible.

8. Completion

The recipient sees the cryptocurrency credited in their wallet. The transaction details (amount, sender, receiver, timestamp, fee, hash) are permanently recorded in the blockchain.

Summary

A cryptocurrency transaction is a secure digital transfer between addresses, cryptographically signed and validated by a decentralized network, then permanently recorded on the blockchain where it cannot be altered or reversed.

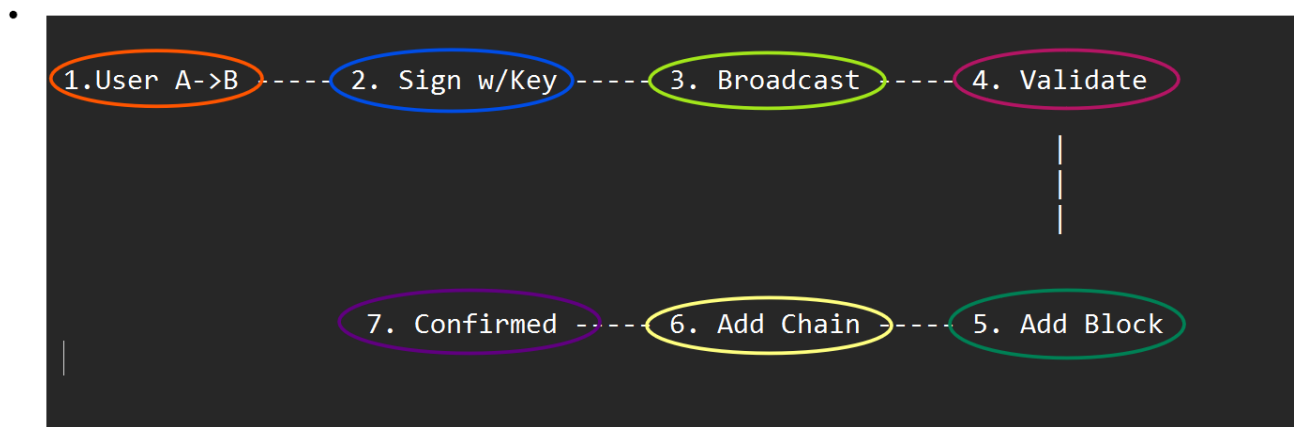


Figure 1.2: How Transaction Works

1.5.3 Types of transaction

1. Transfer Transactions:

1. This is the most basic and common type of transaction.
2. It involves transferring cryptocurrency from one wallet address to another.
3. The transaction records the amount of cryptocurrency sent, the sender's address, and the recipient's address.
4. **Examples** include sending Bitcoin (BTC) from one person to another or transferring Ethereum (ETH) between wallets.
5. These transactions typically include a transaction fee paid to miners or validators to process the transfer.

2. Smart Contract Transactions:

1. These transactions occur on blockchain platforms that support programmable code, such as Ethereum.
2. Instead of simply transferring coins, smart contract transactions execute predefined rules encoded in the contract.
3. Actions include deploying a new smart contract or invoking functions within an existing contract.
4. Examples: decentralized finance (DeFi) operations, automated token trading, lending platforms, and NFTs.
5. Smart contract transactions might involve complex interactions, multiple users, and conditional executions.

3. Token Transactions:

1. Tokens are cryptocurrencies created on top of existing blockchains, such as ERC-20 tokens on Ethereum.
2. Token transactions involve sending or receiving these tokens rather than the blockchain's native coin.
3. These are commonly used for initial coin offerings (ICOs), decentralized applications (dApps), and various utility or governance tokens.
4. Token transfers are recorded on the blockchain, similar to native coin transfers but follow token-specific standards.

4. Mining Reward Transactions (Coinbase Transactions):

1. These transactions create new coins as rewards for miners who successfully validate and add blocks to the blockchain.
2. Mining rewards incentivize miners to maintain network security.
3. Mining transactions do not have inputs because the coins are newly created.
4. They appear as the first transaction in a new block and include the block reward and transaction fees collected by the miner.

5. Contract Creation Transactions:

1. A specialized type of transaction used to create and deploy a new smart contract on the blockchain.
2. This transaction contains the contract code and initial parameters.
3. Once confirmed, the smart contract lives on the blockchain and can be interacted with via other transactions.
4. Contract creation often requires higher fees due to the computational resources needed for deployment.

Block Chain Types and Cryptocurrency transaction:

Let us go through the explanation of broader types of blockchain and cryptocurrency transactions, categorized by the type of blockchain network and transaction context:

1. Public Blockchain Transactions

Description:

Transactions on permissionless blockchains accessible to anyone worldwide.

Examples:

Bitcoin, Ethereum, Litecoin.

Characteristics:

1. Fully transparent and traceable on a public ledger.
2. No central authority controls the network.
3. Transactions require network consensus through mining (Proof of Work) or staking (Proof of Stake).

Examples

Include sending/receiving cryptocurrency and interacting with smart contracts or token transfers.

Use Cases:

Financial transfers, decentralized applications (dApps), decentralized finance (DeFi).

2. Private Blockchain Transactions

Description:

Transactions on permissioned blockchains with restricted access, controlled by a single organization.

Examples

Private consortia blockchains used internally by enterprises.

Characteristics:

1. Limited participation (only authorized entities).
2. Faster transaction processing due to fewer nodes and centralized control.
3. Increased privacy and confidentiality compared to public blockchains.
4. Typically used for internal recordkeeping, supply chain management, or inter-company transactions.

Use Cases:

Enterprise resource planning, internal audits, secure data sharing.

3. Consortium Blockchain Transactions

Description:

Transactions on blockchains governed by a group of organizations rather than a single entity or anyone.

Examples:

Hyperledger Fabric, R3 Corda.

Characteristics:

1. Semi-decentralized, with a defined set of trusted participants.
2. Consensus is achieved among the consortium members.
3. Provides controlled transparency and privacy.
4. Use cases often require collaboration between companies in the same industry.

Use Cases:

Cross-organizational trade finance, healthcare data exchange, syndicated loans.

4. Hybrid Blockchain Transactions

Description:

Combines features of both public and private blockchains, allowing selective access and control.

Examples:

Transactions can be public or private depending on the business rules.

Characteristics:

1. Flexibility to offer transparency to the public while maintaining privacy internally.
2. Enterprises can control who accesses sensitive data.
3. Hybrid approach allows for public accountability with private operations.

Use Cases:

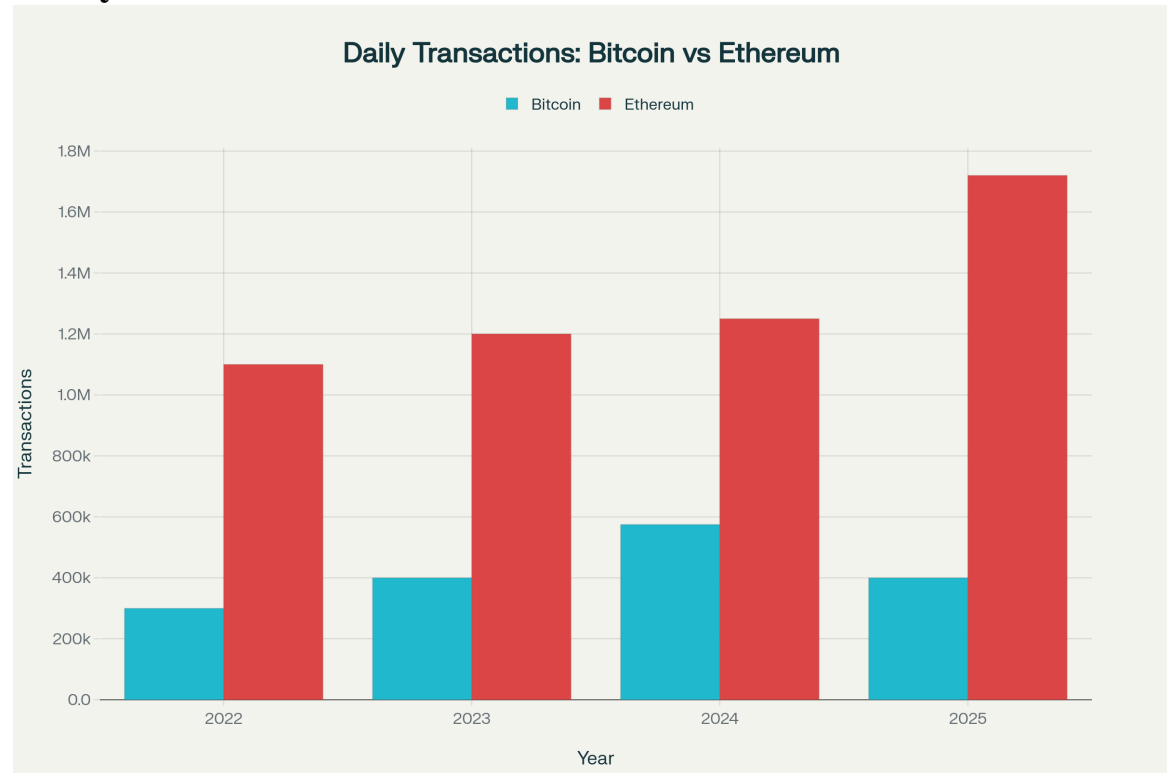
Government record management, private financial transactions with public audits, smart city data management.

Summary

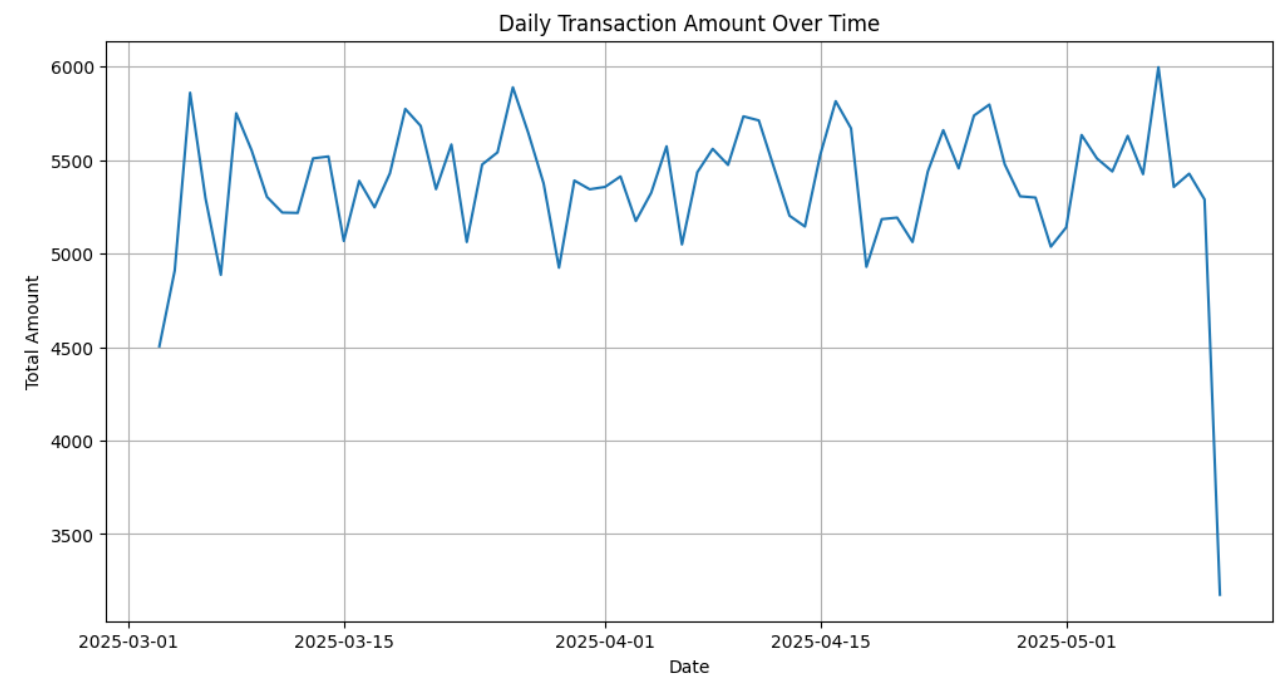
Blockchain Type	Accessibility	Control	Transparency	Use Cases
Public	Permissionless	Decentralized	High	Cryptocurrencies
Private	Permissioned	Centralized	Low	Enterprise applications
Consortium	Permissioned	Selected participants	Partial	Collaborative processes
Hybrid	Both	Combination of both	Variable	Interoperability, governance

BTC and ETH Transaction Analysis

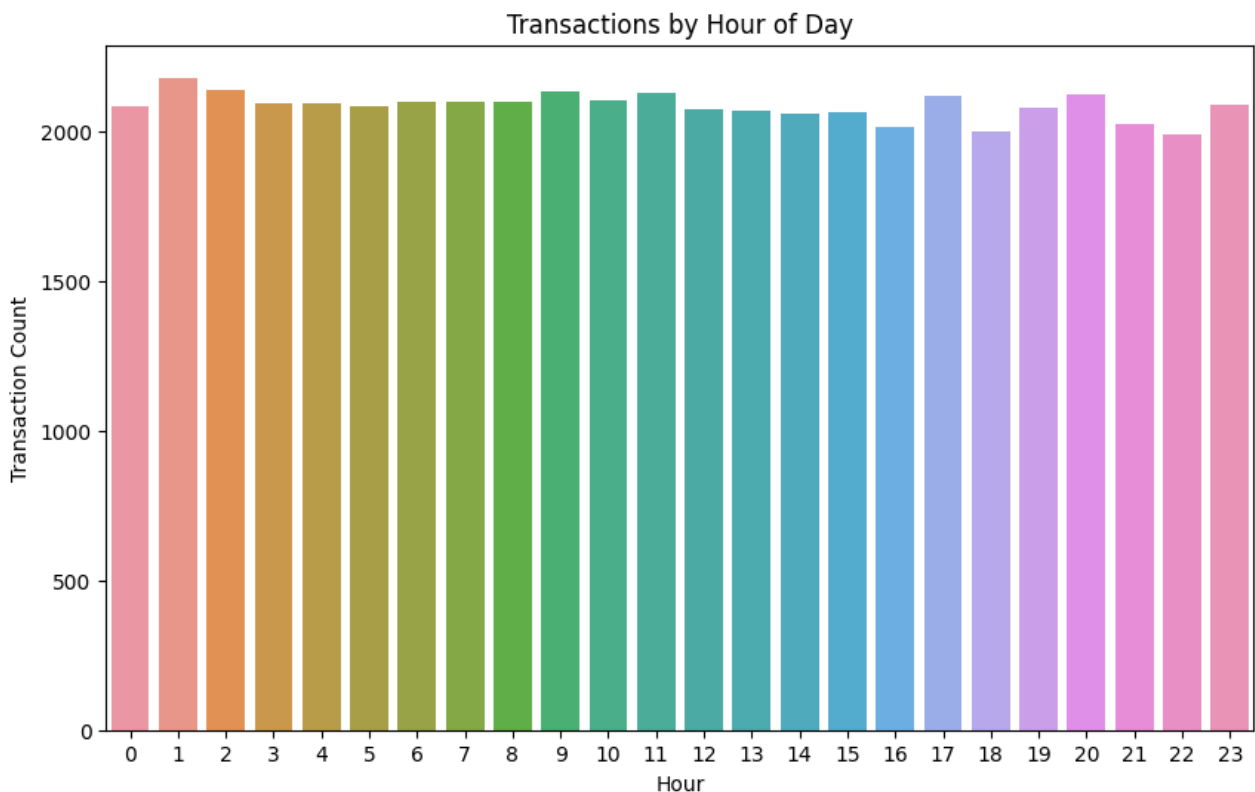
Yearly based:



Monthly based:



Daily based:



Transaction fees:

Cryptocurrency transaction fees are payments made to network validators or miners to process and confirm transactions on a blockchain. These fees serve to incentivize transaction processing and prevent network spam.

Key points about transaction fees in cryptocurrency:

Types of Fees: There are several types including network (miner/validator) fees, trading fees charged by exchanges, withdrawal fees for moving crypto out of exchanges, and deposit fees (less common).

How Fees Work: Fees fluctuate based on blockchain network demand. When the network is congested, fees increase as users compete for limited block space. Users willing to pay higher fees often get faster transaction confirmations.

Fee Calculation: For Bitcoin, fees are typically calculated based on the size of the transaction data in bytes multiplied by a satoshi-per-byte rate. Ethereum uses a gas fee model where each operation consumes gas, and users pay based on gas price and gas used.

Factors Affecting Fees:

1. Network congestion
2. Transaction size or complexity

3. User preferences for transaction speed
4. Blockchain-specific fee structures and governance

Fee Reduction Strategies: Users and institutions may batch transactions, time transactions to off-peak periods, or use fee optimization tools like replace-by-fee (RBF) to reduce costs. Wallet management practices and staking can also help offset fees.

Exchange Fees: Trading on exchanges often involves maker and taker fees based on trading volume, which differ from network transaction fees.

Example Fee Ranges: Bitcoin fees historically ranged from \$0.50 to \$2.50 per transaction but can spike during high demand. Some crypto payment gateways offer transaction fees with varying levels depending on the currency and platform.

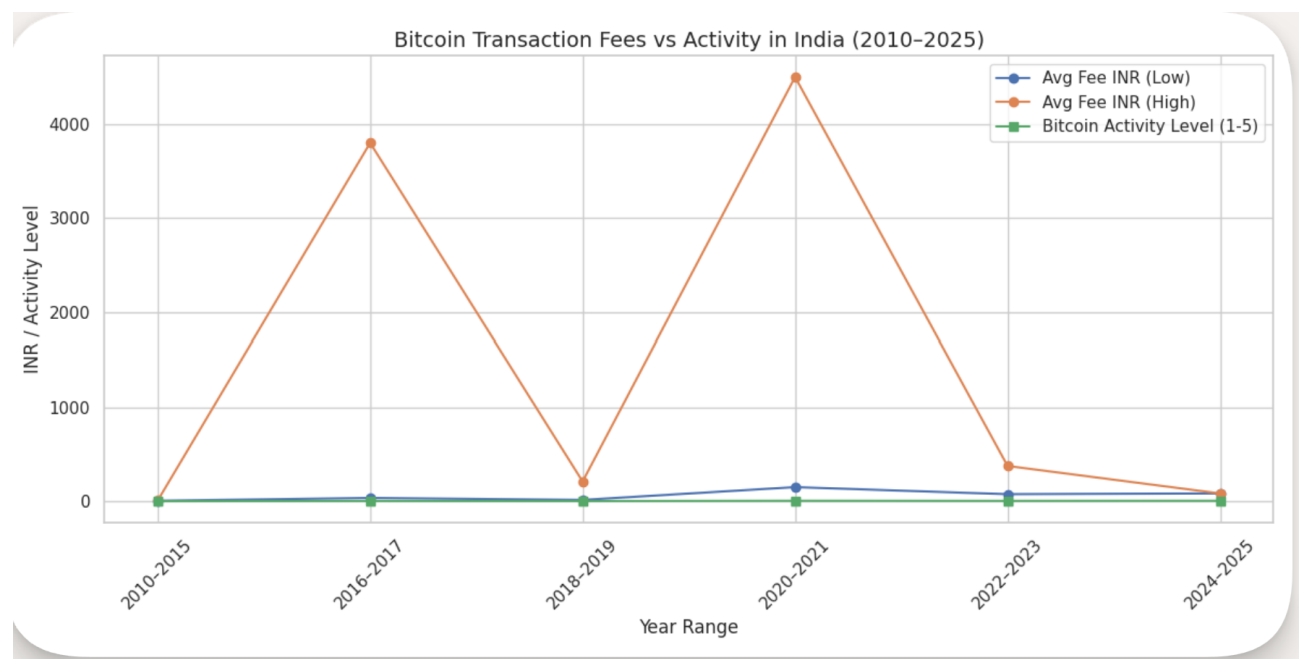


Figure 1.3: transaction fee vs amount

1.6 Cryptocurrency Vulnerabilities:

Cryptocurrency, while revolutionary, faces significant vulnerabilities. These include security flaws in blockchain protocols, susceptibility to hacking, phishing attacks, and wallet breaches. Regulatory gaps and market volatility further expose users to financial risks. Understanding these weaknesses is crucial for safeguarding digital assets and ensuring the integrity of decentralized financial systems .

1.6.1 Common Cryptocurrency Vulnerabilities:

1: Exchange and Wallet Vulnerability: Cryptocurrency exchanges and wallets are frequent targets for hackers due to the large amounts of assets they store. Attacks can include direct hacking, phishing, malware installation, and theft of keys.

2: End-Point Vulnerability: While blockchains themselves are secure due to their immutable ledgers, vulnerabilities often exist at endpoints, such as wallets and interfaces where humans or APIs interact with the blockchain. These are prime targets for theft and manipulation.

3: Phishing and Social Engineering: Cybercriminals trick users into giving up private keys or sensitive information through fake emails, websites, or social media scams. Once the private key is exposed, assets are often permanently lost.

4: Bridge and Smart Contract Attacks: Cross-chain bridges—used to connect different blockchains—are vulnerable to attacks. Weaknesses in smart contract code can also lead to exploits such as re-entrancy and integer overflows (e.g., the DAO hack on Ethereum).

5: 51% Attacks: If a single entity gains more than 50% of a blockchain's mining or staking power, it can manipulate transaction history, double-spend coins, or halt new transactions. Such attacks undermine trust in the affected network.

6: Sybil & Routing Attacks: Attackers can create multiple fake identities to disrupt consensus or intercept network traffic between blockchain nodes, potentially leading to partitioning or double-spending attacks.

7: Malware and Ransomware: Malicious software can infect computers or mobile devices to steal crypto assets or demand ransoms for encrypted data.

1.6.2 Common Crypto Vulnerabilities Targeted by Attackers

1: Blockchain Network Attacks : DDoS, Sybil, routing, eclipse, and long-range attacks.

2: Smart Contract Bugs: Coding errors leading to exploits (e.g., DAO hack).

3: User-Wallet Attacks: Phishing, brute force, key theft, weak signatures.

4: Transaction Verification Attacks: Exploiting consensus (e.g., 51% attacks enabling double spending).

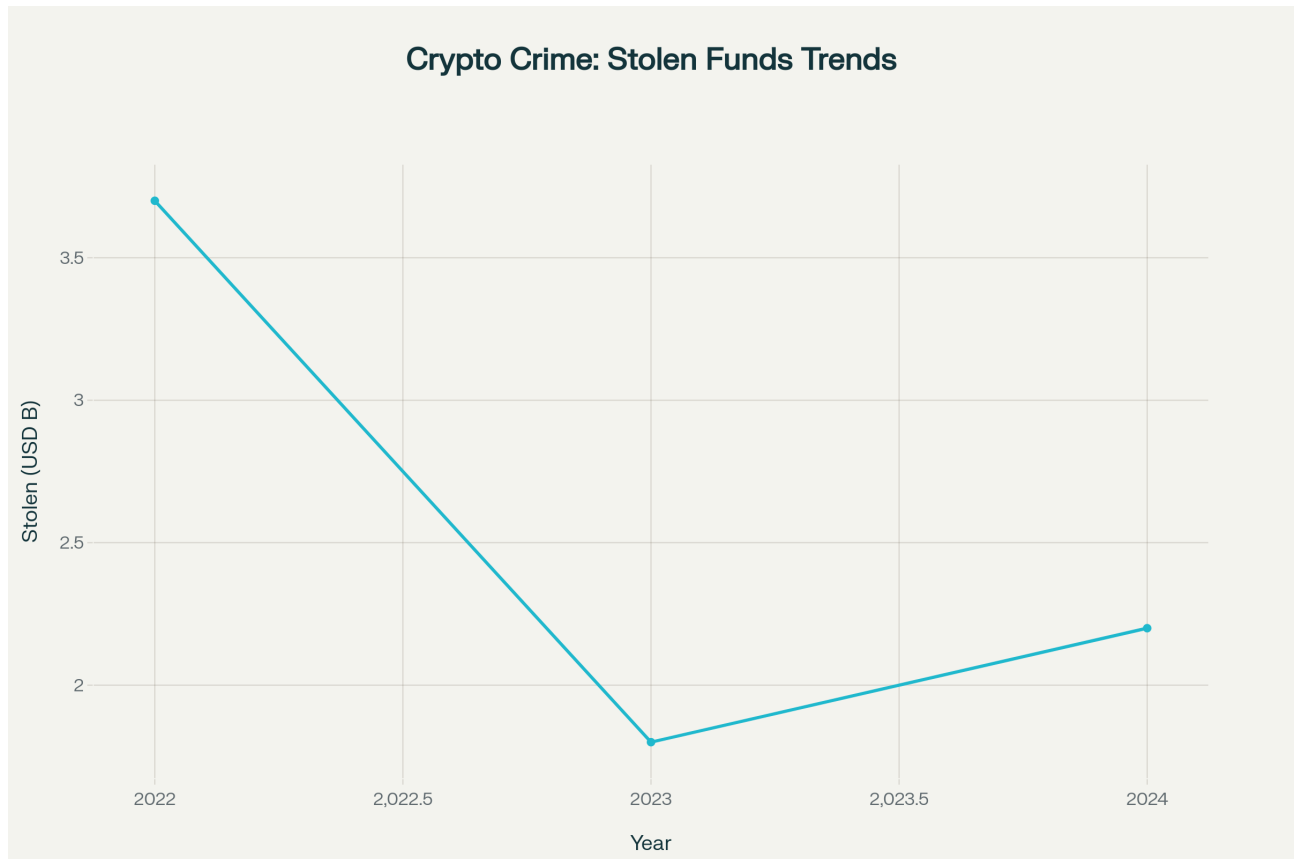


Figure 1.4: Crypto Crimes

1.7 Vulnerability Management in Crypto

Vulnerability management in the crypto space involves a systematic approach to identifying, assessing, prioritizing, and fixing weaknesses that can be exploited by attackers. Here are the key steps and concepts, with practical details:

1. Discover Assets

Begin by identifying all assets within your organization—wallets, nodes, smart contracts, exchanges, APIs, etc.

Full visibility into what you own is critical, as undiscovered assets cannot be protected or assessed.

2. Vulnerability Assessment

Conduct vulnerability scans on your assets, either on-demand or at regular intervals.

Use security tools (e.g., OpenVAS, Acunetix, Intruder) to detect flaws in software, smart contracts, and network configurations.

3. Prioritize Risks

Not all vulnerabilities are equal. Assess risk based on severity, exploitability, and potential business impact.

Focus remediation efforts on vulnerabilities that pose the greatest danger, such as those exposing private keys or allowing unauthorized fund transfers

. 4. Remediate Vulnerabilities

The security/IT team patches or fixes vulnerabilities according to the priority list.

Remediation may include updating smart contracts, securing APIs, tightening access controls, and requiring strong authentication (e.g., 2FA, HSMs, PAM).

Security culture is important at every stage of the software development lifecycle.

5. Measure & Repeat

Track metrics like time-to-remediate, scan frequency, and compliance with industry standards.

Vulnerability management is a continuous cycle, not a one-off exercise. Regular measurement and improvement are essential.

Best Practices for Crypto Vulnerability Management

Penetration Testing: Regularly test systems (including smart contracts) for exploitable bugs before deployment.

1. Secure Coding: Implement industry-standard coding practices and conduct smart contract security audits.

2. Access Control: Restrict wallet and node access to only authorized personnel; use hardware security modules for key storage.

3. Identity & Authorization: Enforce strong authentication, authorization for APIs, and privileged access controls.

4. Data Protection: Use privacy and classification technologies for sensitive data.

5. Regulatory Compliance: Adhere to evolving laws and regulations; vet third-party vendors rigorously.

Common Crypto Vulnerabilities Targeted by Attackers Blockchain Network Attacks: DDoS, Sybil, routing, eclipse, and long-range attacks.

6. Smart Contract Bugs: Coding errors leading to exploits (e.g., DAO hack).

7. User-Wallet Attacks: Phishing, brute force, key theft, weak signatures.

8. Transaction Verification Attacks: Exploiting consensus (e.g., 51% attacks enabling double spending).

1.7.1 Linux tool used:

For vulnerability management specifically in the cryptocurrency and blockchain domain using Kali Linux, focus on these tools and workflows designed to uncover and assess crypto-related vulnerabilities:

Best Kali Linux Tools for Crypto Vulnerability Management

1. OpenVAS (GVM)

Use OpenVAS for scanning blockchain infrastructure (nodes, APIs, wallet servers) against known vulnerabilities.

2. Metasploit Framework

Metasploit can simulate attacks on blockchain nodes, wallet services, and application backends to validate vulnerabilities in crypto infrastructure.

3. Nmap & Nmap Scripting Engine (NSE)

Nmap can discover crypto wallet servers, blockchain nodes, and running services. NSE scripts can identify outdated versions of node software vulnerable to known exploits.

4. Burp Suite & OWASP ZAP

Audit web-based crypto wallets, blockchain explorers, and decentralized app (dApp) frontends for security flaws like XSS, CSRF, and API abuse.

5. Nikto & Wapiti

Scan web servers hosting crypto projects for outdated software and misconfigurations that could expose private keys or backend systems.

6. SQLMap

Test backend databases of crypto exchanges and wallet providers for SQL injection vulnerabilities that could lead to data theft or manipulation.

What Makes Crypto Vulnerability Testing Unique?

Smart Contract Auditing

While Kali tools excel at network, server, and web app testing, smart contracts require specialized auditing. Also consider using tools like Mythril or Slither (not included in Kali by default, but can be installed).

API Security

Crypto platforms use APIs to interact with wallets, exchanges, and blockchains. Kali tools can help fuzz APIs and check for issues like broken authentication and data leaks.

Endpoint Protection

Wallet software and node installations are endpoints; scan them for malware, weak SSL/TLS, or exposed private keys.

Workflow Example

- 1#** Use Nmap to enumerate crypto infrastructure.
- 2#** Apply OpenVAS for full vulnerability scans on the discovered assets.
- 3#** Simulate attacks with Metasploit to test real exploit scenarios.
- 4#** Audit web interfaces and APIs via Burp Suite or OWASP ZAP.
- 5#** Check backend setups with Nikto, Wapiti, and SQLMap.

1.8 Conclusion

The transition to cryptocurrency is reshaping global finance by introducing decentralized, digital assets that offer enhanced transparency, security, and accessibility. Widespread adoption brings opportunities such as financial inclusion, lower transaction costs, and increased privacy for individuals and businesses. However, it also presents challenges—including regulatory uncertainties, scalability limitations, and risks related to volatility and cybersecurity.

A successful transition requires:

- Clear regulatory frameworks to ensure stability and consumer protection.
- Robust technological infrastructure to address scalability and security issues.
- Ongoing education for users and stakeholders to minimize risks and maximize benefits.

Overall, the transition to cryptocurrency represents both promise and caution: it has the capacity to redefine economic interaction, but also demands careful regulation and innovative solutions to realize its full potential.

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